Question 1 (21 marks)

A researcher has asked you for help with this project, below:

A city has introduced solar powered scooters to aid in inner-city transport. Data are collected on 45 consecutive days to determine the relationship between hours of sunlight, called hours, and charge remaining, called life (and also measured in hours). Researchers were interested in predicting charge remaining given hours of sunlight. The data can be found in the file scooters.sav.

a) Explore the relationship between solar hours and life. Include the following:

• A scatterplot with the fitted regression line. (Graph should be labelled appropriately and a line of best fit overlaid).

• A description of the relationship in the sample (2-3 sentences)

(3 marks)

b) Are there any data points that you would recommend that should be deleted and why or why not?

(1 mark)

c) The researcher wants to use all data points and believes the original data will not fit model assumptions. Create the variable in SPSS, log$e$ $hours$. Using linear regression, model the relationship between charge remaining using days and log$e$($hours$). For these relationships, check to see if the fit assumptions of regression have been satisfied.

(3 marks)

d) For the model in (c):

i. write the least squares regression equation in terms of the log$e$($hours$) and charge remaining

ii. interpret the slope in terms of a one unit increase in log hours..

iii. Interpret the constant

iv. predict the life remaining in hours, when the hours of sunlight are 6 hours.

(4 marks)

e) The researcher also wondered how the relationship between charge and sunlight hours changed if the day was sunny or not. The variable sunny is coded 0 for a day that is not sunny and 1 for a day that is sunny. Produce a scatterplot to explore the researcher's hypothesis. Add appropriate regression lines and discuss if there appears to be an interaction present. Describe the interaction or absence of interaction in terms of the variables.

(3 marks)

f) Centre hours of sunlight at its mean and perform a multiple regression analysis to test whether this interaction is statistically significant, including the covariate days in the model. Include the SPSS output. Comment on the findings. (2 marks)

g) Write down the equation of the multiple regression equation obtained in part f. (1 mark)

h) Interpret the constant of this regression (1 mark)

i) Interpret the interaction term of this regression (1 mark)

j) Write down separate regression equations for when it is sunny and when it is not, justifying your answers with SPSS output. (2 mark)

Question 2 (12 marks)

A researcher hypothesized that school stress would be different for children from different family backgrounds. In particular, they hypothesized that children from high conflict families would feel less stress than those from the medium conflict families and that children from medium conflict families would feel less stress than those from low conflict families*.

a) List the assumptions that need to be satisfied to conduct a one-way analysis of variance to address these hypotheses, for each assumption comment whether the assumption has been met. Where possible, verify using SPSS output that the assumptions have been satisfied. (3 marks)

b) Using the data in the file stress.sav, carry out appropriate analyses, and contrasts (3 marks)

c) Write a report which addresses the researcher's hypotheses. (6 marks)

Question 3 (10 marks)

The federal government undertakes a study on productivity of nurses in 3 settings: public hospitals, private hospitals and age care settings. They also considered whether each participant was on day afternoon or night shift and expected to see a difference between productivity as a function of setting. They suspected that productivity would be greatest in the private sector compared to the other sectors, and greatest in the day shift compared to the other shifts and want to test this using contrasts. You have been asked to analyze the data and prepare summary of the results. You do not need to write a report. Use the data contained in nurses.sav to carry out an appropriate analysis of variance to address the management's hypotheses.

There is no need to include a discussion of the normality assumptions for this question or examine post hoc comparisons.